Anmol Gutty

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Education

B.Tech. in Computer Science, Vellore Institute of Technology July 2018 - May 2022
CGPA: 8.59

Experience

Software Engineer, Privafy July 2022 – July 2025

- **Design and Development of Database, APIs, and Workflow for SecTPM:** Spearheaded the design and development of a comprehensive database, APIs, and workflow to support the activation, management, and handling of **SecTPM**.
- **Backend API Architecture for SecEdge-Studio:** Developed robust backend APIs for SecEdge-Studio, focusing on secure account management and scalable user access.
- **Development of Go-Based Services from scratch and CI/CD Implementation:** Engineered four services in Go from the ground up: a mail service for customer communications via SendGrid, a service for device authentication and validation including signature and certification validation, an API server for a new product featuring customer access APIs for device management and certificate generation, and a service for efficient object storage management in GCP buckets(bucket is directly mounted in K8s pod for better efficiency, this reduced memory usage by almost 90% and reduced response times by about 30%). Additionally, developed corresponding CI/CD pipelines using Terraform, Jenkins, and Groovy, including generatiion of their respective certificates.
- **Advanced Server Auditing with ELK Stack:** Enhanced server auditing by integrating the ELK stack for performance analysis and comprehensive log management.
- **Containerization of Scripts:** Containerized Bash and Python scripts to be run directly on Kubernetes pods from inside a Go binary to encrypt and generate firmware and licenses for IoT devices (earlier they were manually run in a VM). This allowed us to integrate it directly into our API flow eliminating the need for manual intervention.
- **TLS Handshake Flow Optimization:** Optimized TLS handshake flow for certificate reusability across devices, reducing storage load by 50% while maintaining robust device identity verification.
- **Cross-Platform Monitoring Tool Development:** Created a monitoring tool for service downtime and performance alerts via email, Microsoft Teams, and Google Chat.
- **Management and Upgrade of Clusters:** Managed upgrades of staging and production clusters, including certificate issuance via Hashicorp Vault, PostgreSQL schema maintenance, and auto-scaling implementation.

Software Engineering Intern, CDAC March 2021 – May 2021

- **Monitoring scripts:** Contributed to the development of robust scripts for monitoring and analyzing system performance in a high performance computing environment.

Skills

Languages: Go, C, C++, Bash/Shell, Python, Java, Kotlin
Databases: PostgreSQL, MongoDB, MySQL
Technologies: ELK (Elasticsearch, Kibana, Logstash), Jenkins, Kubernetes, Docker, GCP, Terraform, Groovy, MQTT, Hashicorp Vault, SendGrid, GNU/Linux
Machine Learning: NLP, Computer Vision, Sentiment Analysis, Tensorflow, Keras, scikit-learn, Matplotlib

Projects

Self hosted server Github

- I've turned an old laptop into a self-hosting server that runs several services: a photo server for backups, a DNS server for ad-blocking, a bookmarking server for organizing links, and a monitoring tool for tracking service usage and uptime. All services are securely accessible worldwide via a mesh VPN.
- Tools Used: Docker, Bash/Shell, Networking, Linux, File management, and various other corresponding tools depending on their use case.

NLP based Movie Recommendation System Github

- Developed and implemented a machine learning system that predicts movie preferences based on users' emotional responses. The system analyzes user reviews to tag movies with relevant emotional categories and combines them with genre and other factors to generate personalized recommendations.
- Tools Used: Sentiment Analysis, NLP, TF–IDF, Tensorflow, Keras, scikit-learn

Pneumonia and Covid Detection

- Developed and integrated separate ensemble models that analyze X-Ray images to accurately detect Covid and pneumonia infections with an impressive accuracy of 88% and 92% respectively. Integrated with a Django-backend based website, enabling users to upload images and receive instantaneous results.
- Tools used: Django, Tensorflow, Keras, AdaBoostClassifier, Matplotlib, MongoDB